Submission: Upload your work to GitHub repo till the next practice lesson

Online Shop

You must start preparing a backend API for your Angular Online Store app from **Lab 5**. This Django backend will eventually serve data to your Angular frontend.

Create a Django project with the name **shop-back**.

Learning Objectives

By the end of this lab, you will be able to:

- Set up a Django project with a virtual environment
- Create Django apps and register them in the project
- Define models with various field types and ForeignKey relationships
- Perform database migrations using makemigrations and migrate
- Build RESTful API endpoints that return JSON responses
- Use URL parameters to filter and retrieve specific data
- Understand how Django processes HTTP requests

Tasks

1. The project contains an application with the name — **api**
2. You have models with the following fields:

Product

- name — CharField
- price — FloatField
- description — TextField
- count — IntegerField
- is_active — BooleanField
- category — ForeignKey to Category

Category

- name — CharField

3. Write API endpoints in JSON format:

Endpoint	Description
/api/products/	List of all Products
/api/products/<int:id>/	Get one Product by ID
/api/categories/	List of all Categories
/api/categories/<int:id>/	Get one Category by ID
/api/categories/<int:id>/products/	List of Products by Category

Requirements

- **Project name:** `shop-back`
- **App name:** `api`
- **Models:** Product and Category with ForeignKey relationship
- **Endpoints:** All 5 API endpoints must return valid JSON
- **Virtual environment:** Must use venv; do NOT push it to GitHub
- **requirements.txt:** Include Django dependency
- **.gitignore:** Exclude venv/, __pycache__/, *.pyc, db.sqlite3

Useful Links

1. **Regular Expressions in URL:** Django URL Dispatcher
2. **How Django Processes a Request:** Request Processing
3. **Configuring the Database:** Database Settings

GOOD LUCK! :)